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# **Derivational Morphology and Vocabulary Expansion**

**DSA0305-Natural Language Processing**

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## Introduction

- Studies the formation of new words.
- Uses prefixes and suffixes.
- Changes meaning or word class.
- Important for vocabulary development.
- Widely used in Natural Language Processing (NLP).



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## **Need For Information Extraction**

- Extracts useful information from large text.
- Saves time and effort.
- Converts unstructured text into structured data.
- Helps find important facts quickly.
- Used in search engines, NLP, and data analysis.



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## COMPONENTS

- Root Word – Basic form of a word.
- Prefix – Added before the root.
- Suffix – Added after the root.
- Derived Word – Newly created word.
- Vocabulary Expansion – Increases the number of words.



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## ADVANTAGES

- Word Formation – Creates new words from existing words.
- Prefixes – Added before a word, e.g., *happy* → *unhappy*.
- Suffixes – Added after a word, e.g., *teach* → *teacher*.
- Meaning Change – Changes or extends the meaning of words.
- Vocabulary Growth – Helps increase and improve vocabulary.



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## **Applications & Conclusion**

- Language Learning – Helps learners understand and build new words.
- Vocabulary Development – Improves vocabulary using prefixes and suffixes.
- NLP Applications – Useful in text analysis and language processing.
- Better Communication – Helps understand word meanings more easily.
- Conclusion – Derivational morphology plays an important role in word formation and vocabulary expansion.